Note

(i) Pole strength is a scalar quantity with dimension $[M^0LT^bA]$. Its SI unit is N T^{-1} (newton per tesla) or A m (ampere-metre).

(ii) Like positive and negative charges in electrostatics, north pole of a magnet experiences a force in the direction of magnetic field while south pole of a magnet experiences force opposite to the magnetic field.

(iii) Pole strength depends on the nature of materials of the magnet, area of cross-section and the state of magnetization.

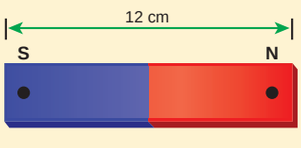
(iv) If a magnet is cut into two equal halves along the length then pole strength is reduced to half.

(v) If a magnet is cut into two equal halves perpendicular to the length, then pole strength remains same.

(vi) If a magnet is cut into two pieces, we will not get separate north and south poles. Instead, we get two magnets. In other words, isolated monopole does not exist in nature.

EXAMPLE 3.3

Compute the magnetic length of a uniform bar magnet if the geometrical length of the magnet is 12 cm. Mark the positions of magnetic pole points.



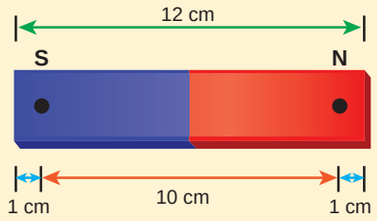
Solution

Geometrical length of the bar magnet is 12 cm

$$Magnetic\ length = \frac{5}{6} \times \left(geometrical\ length \right)$$

$$= \frac{5}{6} \times 12 = 10\,cm$$

In this figure, the dot implies the pole points.



Magnetic field lines

1. Magnetic field lines are continuous closed curves. The direction of magnetic field lines is from North pole to South pole outside the magnet (Figure 3.10) and South pole to North pole inside the magnet.
2. The direction of magnetic field at any point on the curve is known by drawing tangent to the magnetic line of force at that point. In the Figure No. 3.10 (b), the tangent drawn at points P, Q and R gives the direction of magnetic field $\vec{B}$ at that point.
3. Magnetic field lines never intersect each other. Otherwise, the magnetic compass needle would point towards two directions, which is not possible.
4. The degree of closeness of the field lines determines the relative strength of the magnetic field. The magnetic field is strong where magnetic field lines crowd and weak where magnetic field lines thin out.

(d) Magnetic flux

The number of magnetic field lines crossing per unit area is called magnetic flux Φ_B. Mathematically, the magnetic flux through a surface of area A in a uniform magnetic field is defined as

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